Experiment No.: 07

Date:

Experiment Title: State Chart Diagram for Library Management System.

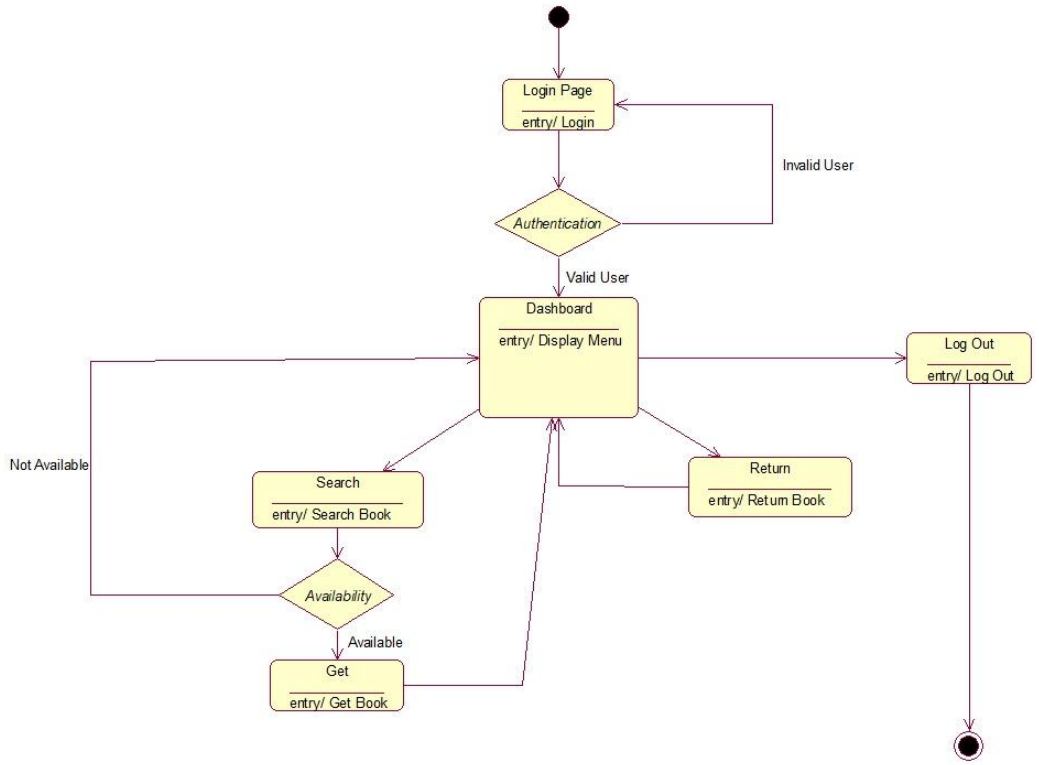
Aim:

To design and understand the State Chart Diagram for a Library Management System that represents the different states and transitions involved in managing library operations.

Problem Description:

A library management system is required to manage activities such as user login, searching for books, checking book availability, issuing books, returning books, and logging out. The system should allow only authenticated users to perform these operations. The state chart diagram is used to represent the behavior of the system by showing different states and transitions between them.

Diagram:



System States Description:

Login Page

This is the initial state where the user enters the username and password to access the system.

Authentication

- The system checks whether the entered credentials are valid.
- If the user is invalid, the system returns to the login page.
- If the user is valid, the system moves to the dashboard.

Dashboard

The dashboard displays the main menu of the system where the user can choose different operations such as searching for books, returning books, or logging out.

Search Book

The user searches for a required book in the library database.

Availability Check

- The system checks whether the searched book is available in the library.
- If the book is not available, the system returns to the dashboard.
- If the book is available, the user can proceed to get the book.

Get Book

The user can borrow or issue the available book from the library.

Return Book

The user returns the borrowed book to the library system.

Log Out

After completing the operations, the user logs out from the system which ends the session.

Result:

The State Chart Diagram for the Library Management System was successfully designed and studied, showing different states such as login, authentication, searching books, checking availability, issuing books, returning books, and logging out.complex real-world problems involving constraints, optimization, and decision-making.